

1 CCSD Equations

Start with the T_1 -transformed equations.

$$E_{CCSD} = \sum_{ia} t_{ia} f_{ia} + \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \langle ij || ab \rangle + \frac{1}{2} \sum_{ijab} t_i^a t_j^b \langle ij || ab \rangle \quad (1)$$

$$R_i^a = \tilde{f}_{ai} + \sum_{me} t_{im}^{ae} \tilde{f}_{me} + \frac{1}{2} \sum_{mef} t_{im}^{ef} \langle am || ef \rangle - \frac{1}{2} \sum_{mne} t_{mn}^{ae} \langle mn || ie \rangle \quad (2)$$

$$R_{ij}^{ab} = \langle ab || ij \rangle + P_{ab}^- \sum_e t_{ij}^{ae} F_{be} - P_{ij}^- \sum_m t_{im}^{ab} F_{mj} + \frac{1}{2} \sum_{mn} t_{mn}^{ab} W_{mnij} + \frac{1}{2} \sum_{ef} t_{ij}^{ef} \langle ab || ef \rangle + P_{ij}^- P_{ab}^- \sum_{me} t_{im}^{ae} W_{mbej} \quad (3)$$

The intermediates are as follows.

$$F_{be} = \tilde{f}_{be} - \frac{1}{2} \sum_{mnf} t_{mn}^{bf} \langle mn || ef \rangle \quad (4)$$

$$F_{mj} = \tilde{f}_{mj} + \frac{1}{2} \sum_{nef} t_{jn}^{ef} \langle mn || ef \rangle \quad (5)$$

$$W_{mnij} = \langle mn || ij \rangle + \frac{1}{2} \sum_{ef} t_{mn}^{ef} \langle mn || ef \rangle \quad (6)$$

$$W_{mbej} = \langle mb || ej \rangle + \frac{1}{2} \sum_{nf} t_{nj}^{fb} \langle mn || ef \rangle \quad (7)$$

1.1 Spin-integrated

$$E_{CCSD} = 2 \sum_{ia} t_i^a f_{ia} + \sum_{ijab} t_{ij}^{ab} L_{ijab} + \sum_{ijab} t_i^a t_j^b L_{ijab} \quad (8)$$

$$R_i^a = 2 \tilde{f}_{ai} + 2 \sum_{me} (2 t_{im}^{ae} - t_{im}^{ea}) \tilde{f}_{me} + 2 \sum_{mef} t_{im}^{ef} \tilde{L}_{amef} - 2 \sum_{mne} t_{mn}^{ae} \tilde{L}_{mnie} \quad (9)$$

$$\begin{aligned} R_{ij}^{ab} = & 2 \langle ab || ij \rangle + 2 P_{ia,jb} \sum_e t_{ij}^{ae} F_{be} - 2 P_{ia,jb} \sum_m t_{im}^{ab} F_{mj} + 2 \sum_{mn} t_{mn}^{ab} W_{mnij} + 2 \sum_{ef} t_{ij}^{ef} \langle ab || ef \rangle \\ & + 2 P_{ia,jb} \sum_{me} (2 t_{im}^{ae} - t_{im}^{ea}) W_{mbej} - P_{ia,jb} \sum_{me} t_{im}^{ea} W_{mbje} - 2 P_{ia,jb} \sum_{me} t_{im}^{eb} W_{maje} \end{aligned} \quad (10)$$

$$F_{be} = \tilde{f}_{be} - \sum_{mnf} t_{mn}^{bf} \tilde{L}_{mnef} \quad (11)$$

$$F_{mj} = \tilde{f}_{mj} + \sum_{nef} t_{jn}^{ef} \tilde{L}_{mnef} \quad (12)$$

$$W_{mnij} = \langle mn || ij \rangle + \sum_{ef} t_{mn}^{ef} \langle mn || ef \rangle \quad (13)$$

$$W_{mbej} = \tilde{L}_{mbej} + \frac{1}{2} \sum_{nf} (2 t_{nj}^{fb} - t_{nj}^{bf}) \tilde{L}_{mnef} \quad (14)$$

$$W_{mbje} = \langle mb || je \rangle - \frac{1}{2} \sum_{nf} t_{jn}^{bf} \langle mn || ef \rangle \quad (15)$$

2 Lambda CCSD

$$R_a^i = \tilde{f}_{ia} - \sum_m \lambda_a^m F_{im} + \sum_e \lambda_e^i F_{ea} + \sum_{me} \lambda_e^m W_{eima} + \frac{1}{2} \sum_{mef} \lambda_{ef}^{im} W_{efam} - \frac{1}{2} \sum_{mne} \lambda_{ae}^{mn} W_{iemn} - \frac{1}{2} \sum_{mn} G_{mn} \langle \widetilde{in||am} \rangle + \frac{1}{2} \sum_{ef} G_{ef} \langle \widetilde{ie||af} \rangle \quad (16)$$

$$R_{ab}^{ij} = \langle \widetilde{ij||ab} \rangle + P_{ij}^- P_{ab}^- \lambda_a^i \tilde{f}_{jb} + P_{ij}^- \sum_e \lambda_e^i \langle \widetilde{ej||ab} \rangle - P_{ab}^- \sum_m \lambda_a^m \langle \widetilde{ij||mb} \rangle + P_{ab}^- \sum_e \lambda_{ae}^{ij} F_{eb} - P_{ij}^- \sum_m \lambda_{ab}^{im} F_{jm} + \frac{1}{2} \sum_{ef} \lambda_{ef}^{ij} W_{efab} + \frac{1}{2} \sum_{mn} \lambda_{ab}^{mn} W_{ijmn} + P_{ij}^- P_{ab}^- \sum_{me} \lambda_{ae}^{im} W_{ejmb} - \frac{1}{2} P_{ab}^- \sum_e G_{be} \langle \widetilde{ij||ae} \rangle - \frac{1}{2} P_{ij}^- \sum_m G_{jm} \langle \widetilde{im||ab} \rangle \quad (17)$$

$$W_{eima} = \langle \widetilde{ei||ma} \rangle + \sum_{nf} t_{mn}^{ef} \langle \widetilde{in||af} \rangle \quad (18)$$

$$W_{efam} = \langle \widetilde{ef||am} \rangle - \sum_n t_{mn}^{fe} \tilde{f}_{na} \quad (19)$$

$$W_{iemn} = \langle \widetilde{ie||mn} \rangle + \sum_f t_{mn}^{fe} \tilde{f}_{if} \quad (20)$$

$$G_{mn} = \sum_{kef} \lambda_{ef}^{mk} t_{nk}^{ef} \quad (21)$$

$$G_{ef} = \sum_{mnc} \lambda_{ec}^{mn} t_{mn}^{fc} \quad (22)$$

$$W_{efab} = \langle \widetilde{ef||ab} \rangle + \frac{1}{2} \sum_{mn} t_{mn}^{ef} \langle \widetilde{mn||ab} \rangle \quad (23)$$

2.1 Spin integration

$$R_a^i = 2\tilde{f}_{ia} - 2 \sum_m \lambda_a^m F_{im} + 2 \sum_e \lambda_e^i F_{ea} + 2 \sum_{me} \lambda_e^m W_{eima} + 2 \sum_{mef} \lambda_{ef}^{im} W_{efam} - 2 \sum_{mne} \lambda_{ae}^{mn} W_{iemn} - 2 \sum_{mn} G_{mn} \tilde{L}_{inam} + 2 \sum_{ef} G_{ef} \tilde{L}_{ieaf} \quad (24)$$

$$R_{ab}^{ij} = 2\langle \widetilde{ij||ab} \rangle + 2P_{ia,jb} \lambda_a^i \tilde{f}_{jb} + 2P_{ia,jb} \sum_e \lambda_e^i \langle \widetilde{ej||ab} \rangle - 2P_{ia,jb} \sum_m \lambda_a^m \langle \widetilde{ij||mb} \rangle + 2P_{ia,jb} \sum_e \lambda_{ae}^{ij} F_{eb} - 2P_{ia,jb} \sum_m \lambda_{ab}^{im} F_{jm} + 2 \sum_{ef} \lambda_{ef}^{ij} W_{efab} + 2 \sum_{mn} \lambda_{ab}^{mn} W_{ijmn} + P_{ia,jb} \sum_{me} (2\lambda_{ae}^{im} - \lambda_{ea}^{im}) W_{ejmb} - P_{ia,jb} \sum_{me} \lambda_{ea}^{im} W_{ejbm} - 2P_{ia,jb} \sum_{me} \lambda_{eb}^{im} W_{ejam} - 2P_{ia,jb} \sum_e G_{be} \langle \widetilde{ij||ae} \rangle - 2 \sum_m G_{jm} \langle \widetilde{im||ab} \rangle \quad (25)$$

$$W_{eima} = \tilde{L}_{eima} + \sum_{nf} (2t_{mn}^{ef} - t_{mn}^{fe}) \tilde{L}_{inaf} \quad (26)$$

$$W_{eiam} = \langle \widetilde{ei||am} \rangle - \sum_{nf} t_{mn}^{fe} \langle \widetilde{in||fa} \rangle \quad (27)$$

$$W_{efam} = \tilde{L}_{efam} - \sum_n (2t_{mn}^{fe} - t_{mn}^{ef}) \tilde{f}_{na} \quad (28)$$

$$W_{iemn} = \tilde{L}_{iemn} + \sum_f (2t_{mn}^{fe} - t_{mn}^{ef}) \tilde{f}_{if} \quad (29)$$

$$G_{mn} = \sum_{kef} (2\lambda_{ef}^{mk} - \lambda_{fe}^{mk}) t_{nk}^{ef} \quad (30)$$

$$G_{ef} = \sum_{mnc} (2\lambda_{ec}^{mn} - \lambda_{ce}^{mn}) t_{mn}^{fc} \quad (31)$$

$$W_{efab} = \langle ef | ab \rangle + \sum_{mn} t_{mn}^{ef} \langle mn | ab \rangle \quad (32)$$

3 Reduced Density Matrix

The one-particle reduced density matrix can be found as follows.

$$\tilde{\rho}_{ij} = \delta_{ij} - \frac{1}{2} \sum_{mef} \lambda_{ef}^{jm} t_{im}^{ef} \quad (33)$$

$$\tilde{\rho}_{ab} = \frac{1}{2} \sum_{mne} \lambda_{ae}^{mn} t_{mn}^{be} \quad (34)$$

$$\tilde{\rho}_{ia} = \lambda_a^i \quad (35)$$

$$\tilde{\rho}_{ai} = \sum_{me} \lambda_e^m t_{im}^{ae} \quad (36)$$

$$\rho_{ij} = \tilde{\rho}_{ij} - \delta_{ij} - \sum_e t_j^e \tilde{\rho}_{ie} \quad (37)$$

$$\rho_{ab} = \tilde{\rho}_{ab} + \sum_m t_m^a \tilde{\rho}_{mb} \quad (38)$$

$$\rho_{ia} = \tilde{\rho}_{ia} \quad (39)$$

In this part of the matrix, the tilde can go over either of the ρ matrices being summed over. The $\tilde{\rho}_{mi}$ is chosen somewhat arbitrarily.

$$\rho_{ai} = \tilde{\rho}_{ai} - \sum_e t_i^e \rho_{ae} + \sum_m t_m^a \tilde{\rho}_{mi} \quad (40)$$

3.1 Spin-integration

The T_1 -transformation equations remain the same, with only one exception. Only the original intermediates change.

$$\tilde{\rho}_{ij} = 2\delta_{ij} - 2 \sum_{mef} (2\lambda_{ef}^{jm} - \lambda_{fe}^{jm}) t_{im}^{ef} \quad (41)$$

$$\tilde{\rho}_{ab} = 2 \sum_{mne} (2\lambda_{ae}^{mn} - \lambda_{ea}^{mn}) t_{mn}^{be} \quad (42)$$

$$\tilde{\rho}_{ia} = 2\lambda_a^i \quad (43)$$

$$\tilde{\rho}_{ai} = 2 \sum_{me} \lambda_m^e (2t_{im}^{ae} - t_{im}^{ea}) \quad (44)$$

$$\rho_{ij} = \tilde{\rho}_{ij} - 2\delta_{ij} - \sum_e t_j^e \tilde{\rho}_{ie} \quad (45)$$

4 Common EOM Matrix Elements

Note that here, W and F are different from what they meant previously.

4.1 One-Particle

At CCSD convergence, this first matrix element becomes zero.

$$\mathcal{F}_{ai} = \tilde{f}_{ai} + \sum_{me} t_{im}^{ae} \tilde{f}_{me} + \frac{1}{2} \sum_{mef} t_{im}^{ef} \langle \widetilde{am} || ef \rangle - \frac{1}{2} \sum_{mne} t_{mn}^{ae} \langle \widetilde{mn} || ie \rangle = 0 \quad (46)$$

$$\mathcal{F}_{ij} = \tilde{f}_{ij} + \frac{1}{2} \sum_{mef} t_{jm}^{ef} \langle im || ef \rangle \quad (47)$$

$$\mathcal{F}_{ab} = \tilde{f}_{ab} - \frac{1}{2} \sum_{mne} t_{mn}^{ae} \langle mn || be \rangle \quad (48)$$

$$\mathcal{F}_{ia} = \tilde{f}_{ia} \quad (49)$$

4.2 Two-Particle

Note the following symmetries. These symmetries allow us to reduce the set of elements to compute, going from 16 to 9.

$$\mathcal{W}_{pqrs} = \mathcal{W}_{qpsr} = -\mathcal{W}_{pqsr} = -\mathcal{W}_{qprs} \quad (50)$$

At CCSD convergence, this first matrix element becomes zero.

$$\mathcal{W}_{abij} = \langle \widetilde{ab} || ij \rangle - P_{ij}^- \sum_m t_{im}^{ab} F_{mj} + P_{ab}^- \sum_m t_{ij}^{ae} F_{be} + \frac{1}{2} \sum_{ef} t_{ij}^{ef} \langle \widetilde{ab} || ef \rangle + \frac{1}{2} \sum_{mn} t_{mn}^{ab} W_{mni j} + P_{ij}^- P_{ab}^- \sum_{me} t_{im}^{ae} W_{mbej} = 0 \quad (51)$$

$$\mathcal{W}_{iajk} = \langle \widetilde{ia} || jk \rangle + \sum_e t_{jk}^{ea} \tilde{f}_{ie} - P_{jk}^- \sum_{me} t_{jm}^{ea} \langle \widetilde{im} || ek \rangle + \frac{1}{2} \sum_{ef} t_{jk}^{ef} \langle \widetilde{ia} || ef \rangle \quad (52)$$

$$\mathcal{W}_{abci} = \langle \widetilde{ab} || ci \rangle - \sum_m t_{mi}^{ab} \tilde{f}_{mc} - P_{ab}^- \sum_{me} t_{mi}^{ae} \langle \widetilde{mb} || ce \rangle + \frac{1}{2} \sum_{mn} t_{mn}^{ab} \langle \widetilde{mn} || ci \rangle \quad (53)$$

$$\mathcal{W}_{ijkl} = \langle \widetilde{ij} || kl \rangle + \frac{1}{2} \sum_{ef} t_{kl}^{ef} \langle ij || ef \rangle \quad (54)$$

$$\mathcal{W}_{abcd} = \langle \widetilde{ab} || cd \rangle + \frac{1}{2} \sum_{mn} t_{mn}^{ab} \langle mn || cd \rangle \quad (55)$$

$$\mathcal{W}_{aijb} = \langle \widetilde{ai} || jb \rangle + \sum_{me} t_{jm}^{ae} \langle mi || eb \rangle \quad (56)$$

$$\mathcal{W}_{ijk a} = \langle \widetilde{ij} || ka \rangle \quad (57)$$

$$\mathcal{W}_{aibc} = \langle \widetilde{ai} | bc \rangle \quad (58)$$

$$\mathcal{W}_{ijab} = \langle \widetilde{ij} | ab \rangle \quad (59)$$

4.3 Spin-integrated forms

$$\mathcal{F}_{ai} = \tilde{f}_{ai} + \sum_{me} (2t_{im}^{ae} - t_{im}^{ea}) \tilde{f}_{me} + \sum_{mef} t_{im}^{ef} \tilde{L}_{amef} - \sum_{mne} t_{mn}^{ae} \tilde{L}_{mnie} = 0 \quad (60)$$

$$\mathcal{F}_{ij} = \tilde{f}_{ij} + \sum_{mef} t_{jm}^{ef} L_{imef} \quad (61)$$

$$\mathcal{F}_{ab} = \tilde{f}_{ab} - \sum_{mne} t_{mn}^{ae} L_{mnbe} \quad (62)$$

$$\mathcal{F}_{ia} = \tilde{f}_{ia} \quad (63)$$

For the spin-integrated matrix elements, the two-body symmetries are fewer. This means we need a couple more matrix elements.

$$\mathcal{W}_{pqrs} = \mathcal{W}_{qpsr} \quad (64)$$

$$\begin{aligned} \mathcal{W}_{abij} = & \langle \widetilde{ab} | ij \rangle - P_{ia,jb} \sum_m t_{im}^{ab} F_{mj} + P_{ia,jb} \sum_m t_{ij}^{ae} F_{be} + \sum_{ef} t_{ij}^{ef} \langle \widetilde{ab} | ef \rangle + \sum_{mn} t_{mn}^{ab} W_{mni j} \\ & + \frac{1}{2} P_{ia,jb} \sum_{me} (2t_{im}^{ae} - t_{im}^{ea}) W'_{mbej} - P_{ia,jb} \sum_{me} t_{im}^{ae} W_{mbje} - P_{ia,jb} \sum_{me} t_{im}^{eb} W_{maje} \neq 0 \end{aligned} \quad (65)$$

$$W'_{mbej} = \langle \widetilde{mb} | ej \rangle + \frac{1}{2} \sum_{nf} (2t_{nj}^{fb} - t_{nj}^{bf}) \langle mn | ef \rangle - \frac{1}{2} \sum_{nf} t_{nj}^{fb} \langle mn | fe \rangle \quad (66)$$

$$\begin{aligned} \mathcal{W}_{iajk} = & \langle \widetilde{ia} | jk \rangle + \sum_e t_{jk}^{ea} \tilde{f}_{ie} - \sum_{me} (2t_{km}^{ae} - t_{km}^{ea}) \langle \widetilde{im} | je \rangle - \sum_{me} t_{km}^{ae} \langle \widetilde{im} | ej \rangle \\ & - \sum_{me} t_{jm}^{ea} \langle \widetilde{im} | ek \rangle + \sum_{ef} t_{jk}^{ef} \langle \widetilde{ia} | ef \rangle \end{aligned} \quad (67)$$

$$\begin{aligned} \mathcal{W}_{abci} = & \langle \widetilde{ab} | ci \rangle - \sum_m t_{mi}^{ab} \tilde{f}_{mc} - \sum_{me} (2t_{mi}^{eb} - t_{mi}^{be}) \langle \widetilde{ma} | ec \rangle - \sum_{me} t_{mi}^{eb} \langle \widetilde{ma} | ce \rangle \\ & - \sum_{me} t_{mi}^{ae} \langle \widetilde{mb} | ce \rangle + \sum_{mn} t_{mn}^{ab} \langle \widetilde{mn} | ci \rangle \end{aligned} \quad (68)$$

$$\mathcal{W}_{ijkl} = \langle \widetilde{ij} | kl \rangle + \sum_{ef} t_{kl}^{ef} \langle ij | ef \rangle \quad (69)$$

$$\mathcal{W}_{abcd} = \langle \widetilde{ab} | cd \rangle + \sum_{mn} t_{mn}^{ab} \langle mn | cd \rangle \quad (70)$$

$$\mathcal{W}_{aijb} = \langle \widetilde{ai} | jb \rangle + \sum_{me} (2t_{jm}^{ae} - t_{jm}^{ea}) \langle mi | eb \rangle - \sum_{me} t_{jm}^{ae} \langle mi | be \rangle \quad (71)$$

$$\mathcal{W}_{aibj} = \langle \widetilde{ai} | bj \rangle - \sum_{me} t_{jm}^{ea} \langle mi | be \rangle \quad (72)$$

$$\mathcal{W}_{ijka} = \langle ij|ka \rangle \quad (73)$$

$$\mathcal{W}_{aibc} = \langle ai|bc \rangle \quad (74)$$

$$\mathcal{W}_{ijab} = \langle ij|ab \rangle \quad (75)$$

5 Right EOM Equations

$$\langle \Phi_i^a | [\bar{H}, \hat{R}] | \Phi_0 \rangle = \sigma_i^a \quad (76)$$

$$\langle \Phi_{ij}^{ab} | [\bar{H}, \hat{R}] | \Phi_0 \rangle = \sigma_{ij}^{ab} \quad (77)$$

$$\sum_e r_i^e \mathcal{F}_{ae} - \sum_m r_m^a \mathcal{F}_{mi} + \sum_{me} r_m^e \mathcal{W}_{amie} + \sum_{me} r_{im}^{ae} \mathcal{F}_{me} + \frac{1}{2} \sum_{mef} r_{im}^{ef} \mathcal{W}_{amef} - \frac{1}{2} \sum_{mne} r_{mn}^{ae} \mathcal{W}_{mnie} = \sigma_i^a \quad (78)$$

$$\begin{aligned} P_{ij}^- \sum_e r_i^e \mathcal{W}_{abej} - P_{ab}^- \sum_m r_m^a \mathcal{W}_{mbij} + P_{ab}^- \sum_e r_{ij}^{ae} \mathcal{F}_{be} - P_{ij}^- \sum_m r_{im}^{ab} \mathcal{F}_{mj} + \frac{1}{2} \sum_{mn} r_{mn}^{ab} \mathcal{W}_{mnij} + \frac{1}{2} \sum_{ef} r_{ij}^{ef} \mathcal{W}_{abef} \\ + P_{ij}^- P_{ab}^- \sum_{me} r_{im}^{ae} \mathcal{W}_{mbej} + P_{ij}^- \sum_n X_{jn} t_{in}^{ab} - P_{ab} \sum_f X_{bf} t_{ij}^{af} = \sigma_{ij}^{ab} \end{aligned} \quad (79)$$

$$X_{jn} = \sum_{mef} r_{jm}^{ef} \langle nm || ef \rangle \quad (80)$$

$$X_{bf} = \sum_{mne} r_{mn}^{be} \langle mn || fe \rangle \quad (81)$$

Once we have found the single and double vector, we can find the reference coefficient.

$$\langle \Phi_0 | [\bar{H}, \hat{R}] | \Phi_0 \rangle = \omega r_0 \quad (82)$$

$$\sum_{ia} r_i^a \mathcal{F}_{ia} + \frac{1}{4} \sum_{ijab} r_{ij}^{ab} \mathcal{W}_{ijab} = \omega r_0 \quad (83)$$

5.1 Spin integration

$$\sum_e r_i^e \mathcal{F}_{ae} - \sum_m r_m^a \mathcal{F}_{mi} + \sum_{me} r_m^e \Omega_{amie} + \sum_{me} (2r_{im}^{ae} - r_{im}^{ea}) \mathcal{F}_{me} + \sum_{mef} r_{im}^{ef} \Omega_{amef} - \sum_{mne} r_{mn}^{ae} \Omega_{mnie} = \sigma_i^a \quad (84)$$

$$\begin{aligned} P_{ia,jb} \sum_e r_i^e \mathcal{W}_{abej} - P_{ia,jb} \sum_m r_m^a \mathcal{W}_{mbij} + P_{ia,jb} \sum_e r_{ij}^{ae} \mathcal{F}_{be} - P_{ia,jb} \sum_m r_{im}^{ab} \mathcal{F}_{mj} + \sum_{mn} r_{mn}^{ab} \mathcal{W}_{mnij} + \\ \sum_{ef} r_{ij}^{ef} \mathcal{W}_{abef} + P_{ia,jb} \sum_{me} (2r_{im}^{ae} - r_{im}^{ea}) \Omega_{mbej} - \frac{1}{2} P_{ia,jb} \sum_{me} r_{im}^{ea} \mathcal{W}_{mbje} \\ - P_{ia,jb} \sum_{me} r_{im}^{eb} \mathcal{W}_{maje} + 2P_{ia,jb} \sum_n X_{jn} t_{in}^{ab} - 2P_{ia,jb} \sum_f X_{bf} t_{ij}^{af} = \sigma_{ij}^{ab} \end{aligned} \quad (85)$$

$$\Omega_{amie} = \Omega_{maei} \quad (86)$$

$$\Omega_{amef} = 2\mathcal{W}_{amef} - \mathcal{W}_{amfe} = \tilde{L}_{amef} \quad (87)$$

$$\Omega_{mnie} = 2\mathcal{W}_{mnie} - \mathcal{W}_{mnei} = \tilde{L}_{mnie} \quad (88)$$

$$\Omega_{mbej} = 2\mathcal{W}_{mbej} - \mathcal{W}_{mbje} = \tilde{L}_{mbej} + \sum_{nf} \left(2t_{jn}^{bf} - t_{jn}^{fb} \right) \tilde{L}_{mnef} \quad (89)$$

$$X_{jn} = \sum_{mef} r_{jm}^{ef} \tilde{L}_{nme} \quad (90)$$

$$X_{bf} = \sum_{mne} r_{mn}^{be} \tilde{L}_{mne} \quad (91)$$

And for the reference coefficient, we get the following.

$$2 \sum_{ia} r_i^a \tilde{f}_{ia} + \sum_{ijab} r_{ij}^{ab} \tilde{L}_{ijab} = \omega r_0 \quad (92)$$